THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL MCQ TEST SET 2

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. A rhombus has perimeter 292 cm and one diagonal 96 cm. Its area (in cm²) is: 1

(a) 2640 **(b)** 7008 **(c)** 5328 **(d)** 5280

2. If $a : b = 3 : 5$ and $b : c = 10 : 7$ and $a + b + c = 322$, then b equals: 1

(a) 84 **(b)** 140 **(c)** 154 **(d)** 98

3. A cube is painted all over and cut into unit cubes. If exactly 24 of the unit cubes have exactly one face painted, the side of the big cube is: 1

(a) 6 **(b)** 4 **(c)** 5 **(d)** 3

4. The mean of the first 12 odd natural numbers is: 1

(a) 6.5 **(b)** 12 **(c)** 144 **(d)** 13

5. If $a^2 - b^2 = 144$ and $a - b = 8$, then $a + b$ equals: 1

(a) 65 **(b)** 18 **(c)** 8 **(d)** 19

6. The area of a triangle is 120 cm². If the altitude on the side of 20 cm is 12 cm, the altitude on the side of 16 cm (in cm) is: 1

(a) 7 **(b)** 17 **(c)** 15 **(d)** 12

7. The number of digits in the square root of a 5-digit perfect square is: 1

(a) 4 **(b)** 2 **(c)** 3 **(d)** 5

8. If $3^x = 243$, then 3^{x-2} equals: 1

(a) 241 **(b)** 27 **(c)** 9 **(d)** 2187

9. The Roman numeral for 1994 is: 1

(a) MCMXCIV **(b)** MMIV **(c)** MCMXCIII **(d)** MCMXCV

10. In an isosceles trapezium ABCD with AB $\parallel$ DC, $\angle A = (3x + 10)^\circ$ and $\angle D = (4x + 30)^\circ$. The measure of $\angle C$ is: 1

(a) 70° **(b)** 130° **(c)** 110° **(d)** 145°

11. If the number 6a2b is divisible by 90, then $a + b$ equals: 1

(a) 1 **(b)** 4 **(c)** 2 **(d)** 3

12. If $(x + p)(x + q) = x^2 + 7x + 12$, then $p^2 + q^2$ equals: 1

(a) 73 **(b)** 49 **(c)** 25 **(d)** 37

13. Two trains cover the same distance at speeds in the ratio 4 : 5. The ratio of the times they take is: 1

(a) 9 : 4 **(b)** 16 : 25 **(c)** 4 : 5 **(d)** 5 : 4

14. The CI on a sum for 2 years at 20% p.a. is ₹ 1100. The sum (in ₹) is: 1

(a) 2250 **(b)** 2750 **(c)** 5000 **(d)** 2500

15. A ladder 13 m long, standing in a street, rests against a house on one side when its top reaches 12 m. Without moving its foot it is turned to rest against the opposite house, reaching 5 m. The width of the street (in m) is: 1

(a) 17 **(b)** 18 **(c)** 19 **(d)** 13

16. 120 men have provisions for 30 days. After 10 days 20 men leave. The remaining provisions last the rest of the men (days): 1

(a) 20 **(b)** 24 **(c)** 26 **(d)** 36

17. A Koch curve starts with a 81 cm line. The length of the curve at stage 4 (in cm) is: 1

(a) 229 **(b)** 324 **(c)** 256 **(d)** 337

18. A data set 54, 34, 23, 16, 31, 46, 57 gets an extra (outlier) value 540. The median of the new data of 8 values is: 1

(a) 34 **(b)** 100.12 **(c)** 40 **(d)** 540

19. The value of $18^2 - 15 \times 21$ is: 1

(a) 18 **(b)** 0 **(c)** 9 **(d)** 6

20. A rhombus has perimeter 116 cm and one diagonal 40 cm. Its area is: 1

(a) 1160 cm² **(b)** 420 cm² **(c)** 840 cm² **(d)** 1680 cm²

21. $\sqrt[3]{-729}$ is: 1

(a) -9 **(b)** -10 **(c)** 10 **(d)** 9

22. The value of $(2^{-1} + 3^{-1})^{-1}$ is: 1

(a) $\frac{1}{5}$ **(b)** $\frac{6}{5}$ **(c)** 6 **(d)** $\frac{5}{6}$

23. The number that comes immediately after 10111_2 in the binary system is: 1

(a) 10110_2 **(b)** 101110_2 **(c)** 11000_2 **(d)** 11001_2

24. The side of a rhombus is $(3x + 2)$ cm and its perimeter is 80 cm. The value of x is: 1

(a) 5 **(b)** 6 **(c)** 8 **(d)** 7

25. The number of steps the Kaprekar routine needs to reach 6174 starting from 2111 is: 1

(a) 4 **(b)** 5 **(c)** 7 **(d)** 6

26. The area of a square of side 25 cm exceeds that of a square of side 22 cm by: 1

(a) 141 **(b)** 9 **(c)** 484 **(d)** 75

27. If 3 : 5 = 12 : 20, then (3+5) : 5 = x : 20, where x is: 1

(a) 12 **(b)** 37 **(c)** 20 **(d)** 32

28. A trader marks his goods 40% above cost and allows a discount of 10%. His gain percent is: 1

(a) 26% **(b)** 29% **(c)** 23% **(d)** 30%

29. A square has a diagonal of $6\sqrt{2}$ cm. Its area (in cm^2) is: 1

(a) 144 **(b)** 72 **(c)** 18 **(d)** 36

30. A vessel has 36 L milk and 12 L water. The litres of water to be added to make the ratio of milk to water 2:1 is: 1

(a) 6 **(b)** 9 **(c)** 24 **(d)** 12

31. A Sierpinski triangle starts with a triangle of side 64 cm. At stage 4 the side of each small shaded triangle is (in cm): 1

(a) 16 **(b)** 8 **(c)** 2 **(d)** 4

32. A car travels 120 km at 40 km/h and then 120 km at 60 km/h. Its average speed (km/h) for the whole journey is: 1

(a) 48 **(b)** 90 **(c)** 24 **(d)** 50

33. The sum of three consecutive integers is 63. The largest of them is: 1

(a) 23 **(b)** 20 **(c)** 22 **(d)** 21

34. A 52 m by 36 m rectangular lawn has a path 4 m wide inside along its boundary. The area of the lawn left is: 1

(a) 1168 m² **(b)** 1808 m² **(c)** 1536 m² **(d)** 1232 m²

35. The ones digit of 127^3 is: 1

(a) 3 **(b)** 7 **(c)** 9 **(d)** 1

36. Which of the following numbers is the greatest? 1

(a) 45×10^4 **(b)** 0.5×10^6 **(c)** 3.2×10^5 **(d)** 480×10^3

37. A number written in base 3 ending in the digit 0 is always divisible by: 1

(a) 10 **(b)** 3 **(c)** 4 **(d)** 2

38. A quadrilateral has equal diagonals that do not bisect each other. It can be an: 1

(a) isosceles trapezium **(b)** parallelogram
(c) rhombus **(d)** kite

39. A number is divisible by both 9 and 11. It must be divisible by: 1

(a) 198 **(b)** 99 **(c)** 20 **(d)** 108

40. The value of $24^2 - 23^2$ is: 1

(a) 24 **(b)** 47 **(c)** 48 **(d)** 49

41. **Assertion (A):** A cube satisfies Euler's formula. 1
Reason (R): For every convex polyhedron, $F + V - E = 2$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

42. **Assertion (A):** The mean of 4, 6, 8, 10 and 12 is 8. 1
Reason (R): The mean of a data set is the sum of the observations divided by the number of observations.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

43. **Assertion (A):** For any whole number n, the number $2n$ is even and $2n + 1$ is odd. 1
Reason (R): Even numbers are multiples of 2, and odd numbers leave remainder 1 when divided by 2.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

44. **Assertion (A):** The area of a parallelogram with base 12 cm and height 5 cm is 60 cm^2. 1
Reason (R): Area of a parallelogram $=$ base $\times$ corresponding height.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

45. **Assertion (A):** The square of an even number is always even. 1
Reason (R): The sum of the first n odd natural numbers is n^2.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

46. **Assertion (A):** $(-1)^{101} = -1$. 1
Reason (R): Any non-zero number raised to the power 0 equals 1.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

47. **Assertion (A):** The Roman numeral XL represents 40. 1
Reason (R): The Roman symbols V, L and D are never repeated.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

48. **Assertion (A):** The diagonals of a rectangle are equal in length. 1
Reason (R): The diagonals of every parallelogram bisect each other.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

49. **Assertion (A):** The number 7,424 is divisible by 4. 1
Reason (R): A number is divisible by 8 if the number formed by its last three digits is divisible by 8.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

50. **Assertion (A):** $(a-b)^2 = (b-a)^2$. 1
Reason (R): $(a+b)(a-b) = a^2 - b^2$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

END OF QUESTION PAPER **Total: 50 Marks**